Turn mixture weights into auditable data exposure

This companion expands notes §7 and blog §7. A mixture is an allocation of token exposures to finite data pools; its weights cannot be interpreted without the corpus inventory and sampling process.

Separate three experiments

A source-addition experiment asks whether adding data improves a chosen training recipe. A mixture experiment searches weights among available buckets. A repetition experiment studies how reuse changes returns. Keep these questions separate initially.

A useful controlled example is Nemotron-CC Table 5: 8B models train for 1T tokens, with 73% from the tested crawl corpus and the same fixed 27% from other sources. DCLM gives MMLU 53.4, full Nemotron-CC 53.0 and the HQ subset 59.0. The HQ result supports that setting, not every longer-horizon regime. The report's full corpus contains 4.4T deduplicated original tokens plus 1.9T synthetic tokens. Su et al., v2, Tables 4–5; Nemotron case study.

Exposure constraints on the simplex

Let U_j be unique tokens in bucket j, w_j its mixture fraction, and D the training-token budget. Then average exposure is $e_j = Dw_j/U_j$, with $w_j \geq 0$ and $\sum_j w_j = 1$. A cap of k_j epochs implies

$$w_j \leq k_j U_j / D.$$

An illustrative 100B-token run assigning 10% to a 1B-token bucket exposes that bucket about ten times. A three-epoch cap allows at most 3% weight. If the sum of all capped capacities is below D, no feasible mixture exists: obtain more unique data, relax justified caps or reduce the horizon.

Per-bucket averages assume the declared sampler. Uneven document sampling can produce a heavy exposure tail. Global overlap between buckets also invalidates treating their token counts as independent capacities.

Preserve exposure in small proxies

Suppose a proxy uses token budget $D_p = sD_t$ relative to a target. To preserve e_j at fixed weights, its available pool must satisfy $U_{jp} = sU_{jt}$. If it instead draws from the whole target pool, it may never encounter the target's repetition problem.

This is an accounting derivation. It does not prove the proxy is a faithful model of target generalization. Confirm mixture rankings at a larger scale, with architecture, optimizer and target exposures matched. Marin data pipeline; Olmix.

Understand the saturation formula

For $D \geq U$, let $R = D/U - 1$ count repeats beyond the first epoch. One fitted effective-data model is

$$D_{\text{eff}} = U[1 + R_*(1 - e^{-R/R_*})].$$

Its derivative with respect to D is e^{-R/R_*}, so the marginal effective contribution decreases exponentially. Its limit is $U(1 + R_*)$. Muennighoff et al.'s Appendix A reports $R_D^* \approx 15.4$ for its fit; this is an e-folding scale, not four epochs and not a half-life. Data-constrained scaling, v5, Appendix A.

The monotone saturation form cannot describe validation loss getting worse from severe overfitting. Diagnose such failures rather than forcing every point into it. Synthetic rewrites also need lineage: different wording is not independent new source knowledge.

Select and confirm a mixture

Fit task-specific responses on sampled feasible mixtures, validate on excluded mixtures, solve under exposure constraints, and compare the solved mixture with natural and capped baselines. A compositional correlation is not a causal effect: raising one weight necessarily lowers others. Preserve the substitution being tested.

Required artifact: bucket inventory, overlap audit, sampler definition, exposure table, held-out mixture scores and larger-scale confirmation. See the data protocol.

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